RESEARCH SUMMARY

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1. Introduction

I am a geometric topologist studying dynamical structures in 3-manifolds and geometric group theory. Thurston pioneered techniques that bridged low dimensional topology, hyperbolic geometry, and dynamical systems—a common motif is simplifying the dynamics by introducing lower-dimensional structures carrying those same dynamics. In my work, I focus on understanding these constructions "from infinity;" rather than studying the specific behavior of a dynamical system, I consider its asymptotic behavior. This asymptotic behavior is often extremely rigid even when the dynamics that produce it are flexible. I use the large-scale geometry of these dynamical systems to understand the manifolds and groups supporting them.

The prototypical example of this simplification technique is that of a ***universal circle***, a circle S^1_u together with an action of the fundamental group $\pi_1 M \curvearrowright S^1_u$ encoding the dynamics. These circles are associated to many different dynamical structures and distinct structures on the same manifold will produce different universal circles. Universal circles can usually (and conjecturally always) be studied via $\pi_1 M$-equivariant ***sphere filling curves*** $S^1_u \to S^2$ which contain additional dynamical information. In the group theoretic setting, these same techniques unlock new descriptions of hyperbolic groups and their splittings. The bulk of my work has been focused on two fronts:

(1) In work with my advisor Steven Frankel, we show that any sphere filling curve $S^1 \to S^2$ approximated by embeddings has the same rich structure as those coming from 3-manifold constructions. [FK26].
(2) In work with Maxwell Plummer, we construct and classify the universal Cantor sets of a free-by-cyclic group in a way that parallels how the Thurston norm organizes universal circles. [KP26]

In the future, I plan to build on this work and:

(3) Use (1) to construct 3-manifolds from sphere-filling curves.
(4) Use (1) to construct new sphere-filling curves associated to dynamical structures which are not known to admit them.
(5) Compare the maps produced in (2) to one another to better understand the various splittings of a free-by-cyclic group.
(6) Generalize the results of (2) to topological structures in these groups that do *not* come from group-theoretic splittings.

2. Motivation: Fibered 3-manifolds and Cannon-Thurston Maps

We now cover the motivating example for much of my work—the first of these $\pi_1 M$-equivariant sphere filling curves constructed by Cannon and Thurston in [CT07]. Here and throughout all manifolds are closed, hyperbolic 3-manifolds and a ***flow*** on M is a continuous action $\mathbb{R} \curvearrowright M$ without global fixed points. A ***foliation*** on a 3-manifold is a decomposition of M into surfaces while a foliation on a plane is a decomposition into lines and pronged singularities—both types of foliations require that the surfaces/lines, called ***leaves*** of the foliation, fit together with a local product structure.

Suppose that $f : \Sigma \to \Sigma$ is a homeomorphism of a closed surface with genus at least two. Using f, we may create a 3-manifold as follows: First, observe that $\Sigma \times [0,1]$ is a compact 3-manifold with boundary given by two copies of Σ. We then create M_f, the ***mapping torus*** of f, by gluing one boundary component to the other along f:

$$M_f = \Sigma \times [0,1]/(p,0) \sim (f(p),1)$$

Such an M_f is a closed 3-manifold with a fibration $M_f \to S^1$ whose point-preimages are each a copy of Σ. Further, the dynamics of f determines the geometric structure of M_f. A celebrated theorem of Thurston is that M_f is hyperbolic (a geometric property) if and only if no power of f fixes the isotopy class of any simple closed curve on Σ (a topological property) [Thu88]. Homeomorphisms with this property are called ***pseudo-Anosov***.

M_f comes with a ***suspension flow*** φ given by flowing in the 'vertical' direction which is transverse to the ***foliation by fibers***, the image of $\{\Sigma \times \{x\} \mid x \in [0,1]\}$ under the quotient map. We construct the ***orbit space*** of φ by lifting φ to a flow $\tilde{\varphi}$ on the universal cover $\tilde{M}$ and quotienting each flow line of $\tilde{\varphi}$ to a point: $\mathcal{O}_\varphi = \tilde{M}/\tilde{\varphi}$. The orbit space $\mathcal{O}_\varphi$ is the prototypical example of a ***bi-foliated plane***—a plane P supporting a pair of foliations which meet transversely at every point.

Giving Σ and M_f their natural hyperbolic structures, we may identify their universal covers with hyperbolic 2- and 3-space respectively: as such, the inclusion $\Sigma \hookrightarrow M_f$ lifts to a map $\mathbb{H}^2 \to \mathbb{H}^3$. A celebrated theorem of Cannon and Thurston [CT07] is that this map extends continuously to their natural compactifications $\mathbb{H}^2 \sqcup S^1_\infty \to \mathbb{H}^3 \sqcup S^2_\infty$. Restricting our attention to the boundaries, the ***Cannon-Thurston map*** $CT : S^1_\infty \to S^2_\infty$ is $\pi_1 M$ equivariant. As the action $\pi_1 M \curvearrowright S^2_\infty$ is minimal, the image is dense—thus CT is surjective.

These results have two main directions for generalization. Staying in the world of 3-manifolds we can build additional topological or dynamical structures and construct their universal circles. On the other hand, Cannon and Thurston's construction has a natural group theoretic analogue—building the geometric group theory of certain hyperbolic groups fibering over the circle. In both cases, it is the Cannon-Thurston map and its generalizations that provides the link between the original and simplified dynamics.

2.1. **Overview.**

My research program can be broadly divided into two subfields and I devote a section below to each of them. *Cannon-Thurston Maps onto S^2* covers the work done in [FK26] in the setting of hyperbolic 3-manifolds. *Cannon-Thurston Maps for Free-by-Cyclic Groups* covers the work done in [KP26] generalizing well-known results in 3-manifolds to a new setting.

3. Cannon-Thurston Maps onto S^2

A flow ψ on a 3-manifold M is a ***pseudo-Anosov flow*** if it is locally modeled on the suspension flow of a pseudo-Anosov map. More generally, a flow φ on M is ***quasigeodesic*** if, after lifting φ to a flow $\tilde{\varphi}$ on $\tilde{M}$, each flowline of $\tilde{\varphi}$ stays a bounded distance from a geodesic—pseudo-Anosov flows have incredibly rigid local structure while quasigeodesic flows are more flexible. Recent work of Frankel and Landry shows that any quasigeodesic flow on a closed, hyperbolic 3-manifold can be deformed into a pseudo-Anosov one. The orbit space of a pseudo-Anosov flow $\mathcal{O}$ is topologically a plane with a pair of foliations determined by the dynamics of the flow—the same is true for quasigeodesic flows but the foliations are instead a pair of decomposition [Cal06; FM01]. This orbit space has a canonical compactification as a closed disk $\mathcal{O} \sqcup S^1_u$ whose boundary circle is a universal circle for M with a Cannon-Thurston map $CT : S^1_u \to S^2$ [Cal06; Fen12; Fra15]. The action $\pi_1 M \curvearrowright S^1_u$ fixes a pair of ***laminations***—a closed set of unordered pair of points in S^1; these laminations $\Lambda^{\pm}$ provide data about how the Cannon-Thurston Map identifies points on S^1_u. In both the fibered setting and this flow setting, a structure on M produces the data $(\mathcal{O}, S^1_u, \Lambda^{\pm}, CT)$. My work with Frankel provides a partial converse—we can reproduce this data from the Cannon-Thurston Map alone:

Theorem 3.1 (Frankel-K. [FK26])**.** *Suppose that $f_\infty \colon S^1 \to S^2$ is a sphere filling curve and that there is a continuous family $f_t : S^1 \to S^2$, $t \in [0, \infty]$, such that f_t is an embedding when $t < \infty$. Then f_∞ supports a pair of laminations $\Lambda^{\pm}$ extending to bifoliated plane.*

Previous constructions linking Cannon-Thurston maps and bifoliated planes use the geometry of $\mathbb{H}^3$ as an integral component. Our proof of Theorem 3.1 is purely topological; we use the asymptotic behavior of the approximations f_t to build our laminations and the laminations reconstruct the bifoliated plane by [FL25].

Another example of a universal circle is the ***leftmost circle*** $\mathcal{C}^\ell$ of an ***Anosov flow*** on a hyperbolic 3-manifold—a flow is Anosov if, around every flow line, the flow expands in one direction and contracts in the other. An Anosov flow determines a pair of invariant foliations on M called the ***stable*** and ***unstable foliations***. After lifting to the universal cover, a leaf in the stable (unstable) foliation is a set of flow lines which are forward (backward) asymptotic to the same point in $\partial_\infty \tilde{M}$. The leftmost circle describes how the asymptotic behavior changes as one moves from between leaves. Recent work by Buckminster [Buc26] produces a Cannon-Thurston map for $\mathcal{C}^{\ell/r}$, inspiring the following question:

Question 1. *Suppose that S^1_u is a universal circle for M. Does there exist a Cannon-Thurston map $\varphi : S^1_u \to S^2_\infty$?*

While universal circles arising from orbit spaces come with the additional data in the form of a pair of circle laminations and corresponding bi-foliated plane, this is not always the case—$\mathcal{C}^{\ell/r}$ lack this structure. Conjecturally, every universal circle is the boundary of the orbit space of some pseudo-Anosov flow [LMT24].

Question 2. *Suppose that S^1_u is a universal circle for M admitting a Cannon-Thurston map $\varphi : S^1_u \to S^2_\infty$. Does φ admit a pair of laminations extending to a bifoliated plane?*

My work with Frankel opens a new angle of attack for this question. Theorem 3.1 reduces Question 2 to showing that the Cannon-Thurston maps are approximated by embeddings.

Further, the construction in Theorem 3.1 is natural in the sense that if $f_\infty : S^1 \to S^2$ is equivariant under the action of some group G, then the action of G in S^1 preserves the laminations $\Lambda^\pm$ and G acts on the corresponding bifoliated plane.

This work is a part of a larger program by Calegari to create a dictionary translating between geometric and dynamical structures on 3-manifolds and topological structures on S^2. Calegari and Loukidou introduce ***CaTherine Wheels*** and ***Zippers*** [CL24; CL26] as the start of this dictionary. CaTherine wheels are a class of sphere filling curves $f\colon S^1 \to S^2$ such that for every closed interval $I \subset S^1$, $f(I)$ is a closed disk and $f(\partial I) \subset \partial f(I)$. All such maps are approximated by embeddings and Theorem 3.1 is a key step in developing their dictionary. Importantly however, there are many topological and dynamical structures whose Cannon-Thurston maps are *not* CaTherine Wheels—but they *are* approximated by embeddings. Zippers are a pair of disjoint ***real trees*** in $Z^\pm \subset S^2$ which are invariant under the action of $\pi_1 M$—the prototypical example is the forward and backward endpoints of a quasigeodesic flow. In the same way that CaTherine Wheels are approximated by embeddings, zippers are approximated by simplicial trees; in fact, given an approximation C of a CaTherine Wheel, one can choose approximations of the zippers so that each the approximated zippers live on either side of the Jordan curve C. A major future direction for my research is continue to develop this dictionary and to build 3-manifolds from their Cannon-Thurston Maps:

Conjecture 3.2. *Suppose G acts as a uniform convergence group on S^2 and that $f\colon S^1 \to S^2$ is a G-invariant sphere filling curve approximated by embeddings. Then there is a closed, hyperbolic 3-manifold M and a quasigeodesic flow φ on M such that $G = \pi_1 M$ and f is the Cannon-Thurston map for φ.*

4. Cannon-Thurston Maps for Free-by-Cyclic Groups

A celebrated result of Mj says that whenever

$$1 \to H \to G \to Q \to 1$$

is a short exact sequence of infinite word hyperbolic groups, there is a surjective Cannon-Thurston map $\partial_\infty H \to \partial_\infty G$ between the Gromov boundaries of H and G [Mit98]. Cannon and Thurston's result is then the case where $H = \pi_1 \Sigma$, $G = \pi_1 M$, and $Q = \mathbb{Z}$; notice that we can write G as the semidirect product $\pi_1 M \simeq \pi_1 \Sigma \rtimes_{f_*} \mathbb{Z}$ where $f_*\colon \pi_1 \Sigma \to \pi_1 \Sigma$ is induced by gluing map $f : \Sigma \to \Sigma$.

Let F_N denote the free group on N letters; given an outer automorphism $\varphi \in \mathrm{Out}(F_N)$, we may form the ***free-by-cyclic*** $G \simeq F_N \rtimes_\varphi \mathbb{Z}$. As long φ satisfies certain technical conditions related to how it mixes free factors in F_N (fully irreducible and atoroidal), G is a hyperbolic group that shares many properties with a fibered hyperbolic 3-manifold. By the aforementioned result of Mj, there is a Cannon-Thurston map $\partial_\infty F_N \to \partial_\infty G$. Unlike the fibered setting, these spaces at infinity are less well-behaved: if $N \geq 3$, then $\partial_\infty F_N$ is a Cantor set and $\partial_\infty G$ is the Menger Sponge [KK00].

While a hyperbolic free-by-cyclic group $G \simeq F_N \rtimes_\varphi \mathbb{Z}$ is a purely group theoretic construction, there are many ways to reintroduce topology. An ***expanding train track map*** for φ is a graph Γ with smoothing data around each vertex and a homotopy equivalence $f\colon \Gamma \to \Gamma$ inducing φ on $\pi_1 \Gamma$. Dowall-Kapovich-Leininger use the data of an expanding train track map

to construct the ***folding mapping torus***, a $K(G,1)$ with a fibration $X_f \to S^1$ whose point preimages are graphs homotopy equivalent to Γ and monodromy f [DKL15]. The folded mapping torus comes equipped with the ***suspension semi-flow*** ψ_t (an action $\mathbb{R}_+ \curvearrowright X_f$) defined analogously to the suspension flow on a mapping torus; as f is not invertible, we can only flow forwards, hence the restriction to a semi-flow.

The first natural question about a hyperbolic free-by-cyclic group is whether the splitting is unique. In general, the answer is a resounding no—most free-by-cyclic groups split infinitely many ways with infinitely many different monodromies. The splittings of G are governed by the ***symmetric BNS invariant***, a finite union of open cones $\Sigma_s(G) \subset H^1(G;\mathbb{R})$ [BNS87]. A surjective homomorphism $u : G \to \mathbb{Z}$ satisfies $[u] \in \Sigma_s(G)$ if and only if $\ker u$ is finitely generated; in this case, $\ker u$ is free and we can write $G \simeq \ker u \rtimes \mathbb{Z}$ [GMSW01]. The folded mapping torus is an invariant of an entire component of the BNS invariant. If $[u]$ and $[v]$ belong to the same component of the BNS invariant with monodromies φ, ψ and expanding train track maps $f : \Gamma \to \Gamma$ and $g : \Theta \to \Theta$, then there is a homeomorphism $X_f \to X_g$ between their folded mapping tori sending semiflow lines to semiflow lines and inducing an isomorphism between the splittings $F_N \rtimes_\phi \mathbb{Z} \simeq G \simeq F_{N'} \rtimes_\psi \mathbb{Z}$. The utility of the BNS invariant motivates:

Question 3. *Given a splitting of a free-by-cyclic group $G \simeq F_N \rtimes_\varphi \mathbb{Z}$, what properties of the splitting depend only on the group G? Which properties depend only on the component of the BNS Invariant?*

Question 3 has many partial answers. For example, Mutanguha proved that (full) irreducibility of φ is an invariant of the group [Mut21], building on work of Dowall–Kapovich–Leininger, who showed it was an invariant of a component of the BNS invariant [DKL17]. Maxwell Plummer and I answer Question 3 for the Cannon-Thurston Map.

Theorem 4.1 (K.-Plummer [KP26])**.** *Let $G \simeq F_N \rtimes_\varphi \mathbb{Z}$ be a hyperbolic free-by-cyclic group with fully irreducible monodromy. The Cannon-Thurston map $\partial_\infty F_N \to \partial_\infty G$ depends only on the component of the BNS Invariant.*

We construct a ***Universal Cantor*** set $\mathfrak{C}$ and a map $\mathcal{R} : \mathfrak{C} \to \partial_\infty G$ such that for any $u : G \to \mathbb{Z}$ which belongs to the same component as φ there is an identification $h_u \colon \partial_\infty \ker u \to \mathfrak{C}$ such that the Cannon-Thurston map $CT_u : \ker u \to \partial_\infty G$ factors as $CT_u = \mathcal{R} \circ h_u$.

Building on this result, I hope to compare the universal Cantor sets across different components of the BNS Invariant. In the world of 3-manifolds, universal circles from different components of the BNS Invariant for $\pi_1 M$ are distinct in both their $\pi_1 M$ actions and their Cannon-Thurston maps. Thus, I hope to prove

Conjecture 4.2. *Suppose that G is a hyperbolic free-by-cyclic group with fully irreducible monodromy and let $\mathfrak{C}$ and $\mathfrak{C}'$ denote the universal Cantor sets associated to distinct component of $\Sigma_s(G)$. Then the actions $G \curvearrowright \mathfrak{C}$ and $G \curvearrowright \mathfrak{C}'$ are not conjugate.*

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